Energia $\quad \bar{r} \simeq \text{cost} \quad \Rightarrow \quad E_{pot} = E(\bar{r}) = \text{cost}$

$$\Rightarrow \Delta E_m = \Delta E_k$$

Urti $\begin{cases} \text{elastici} & \boxed{\Delta E_k = 0} \quad (\text{forze conservative}) \\ \text{anelastici} & \Delta E_k \neq 0 \quad (\text{forze dissipative}) \end{cases}$

Sistema isolato $\quad \bar{R}^E_i = 0 \qquad \bar{R}^E = \dfrac{d\bar{P}}{dt}$

$$\Rightarrow \boxed{\bar{P}(t_0^-) = \bar{P}(t_0^+)}$$

$\bar{R}^E_i \neq 0$ forze <u>non</u> impulsive $\Rightarrow \underset{\substack{\wr\wr \\ 0}}{|\Delta \bar{P}^E|} \ll |\Delta \bar{P}^I|$

Sistema isolato $\Rightarrow \bar{M}^E_{0,i} = 0$

$$\bar{M}^E_0 = \frac{d\bar{L}_0}{dt} - \bar{v}_0 \times m\bar{v}_{CM}$$

polo $\quad v_0 = 0 \;, \; O \equiv CM \quad \Rightarrow \quad \bar{M}^E_{polo} = \dfrac{d\bar{L}_{polo}}{dt} = 0$

$$\Rightarrow \boxed{\bar{L}_{polo} = cost} \quad \Rightarrow \boxed{\bar{L}_{polo}(t_0^-) = \bar{L}_{polo}(t_0^+)}$$

Sistema non isolato

$\bar{M}^E \neq 0$ non impulsivi $\Rightarrow \underbrace{|\Delta \bar{L}^E_{polo}|}_{\simeq 0} << |\Delta \bar{L}^I_{polo}|$

$\bar{M}^E \neq 0$ impulsivo $\Rightarrow \boxed{\bar{L} \neq cost}$